Contextro: How We Built a Local Code Intelligence MCP That Cuts Agent Token Usage by 98%

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TL;DR

We built Contextro, an open-source MCP server that gives AI coding agents a brain. Instead of reading files and guessing, agents search by meaning, trace call graphs, and check what breaks — all locally, in milliseconds.

We ran controlled experiments on a production monorepo (8,496 files). The results:

- **98.3% fewer tokens** consumed per workflow
- **81× faster** discovery (5.8ms vs 471ms median)
- **Zero files read** — agents never need to open a file to find something
- **35 tools**, all completing successfully under 500ms (most under 10ms)

This post explains what we built, how we achieved this performance, and the research methodology behind it.

The Problem

AI coding agents are powerful but wasteful. When Claude, Cursor, or Copilot needs to find a function, it reads 3–5 files hoping to stumble on the right one. Each file costs 500–2,000 tokens. A simple "find this function" task burns 5,000+ tokens before the agent even starts working.

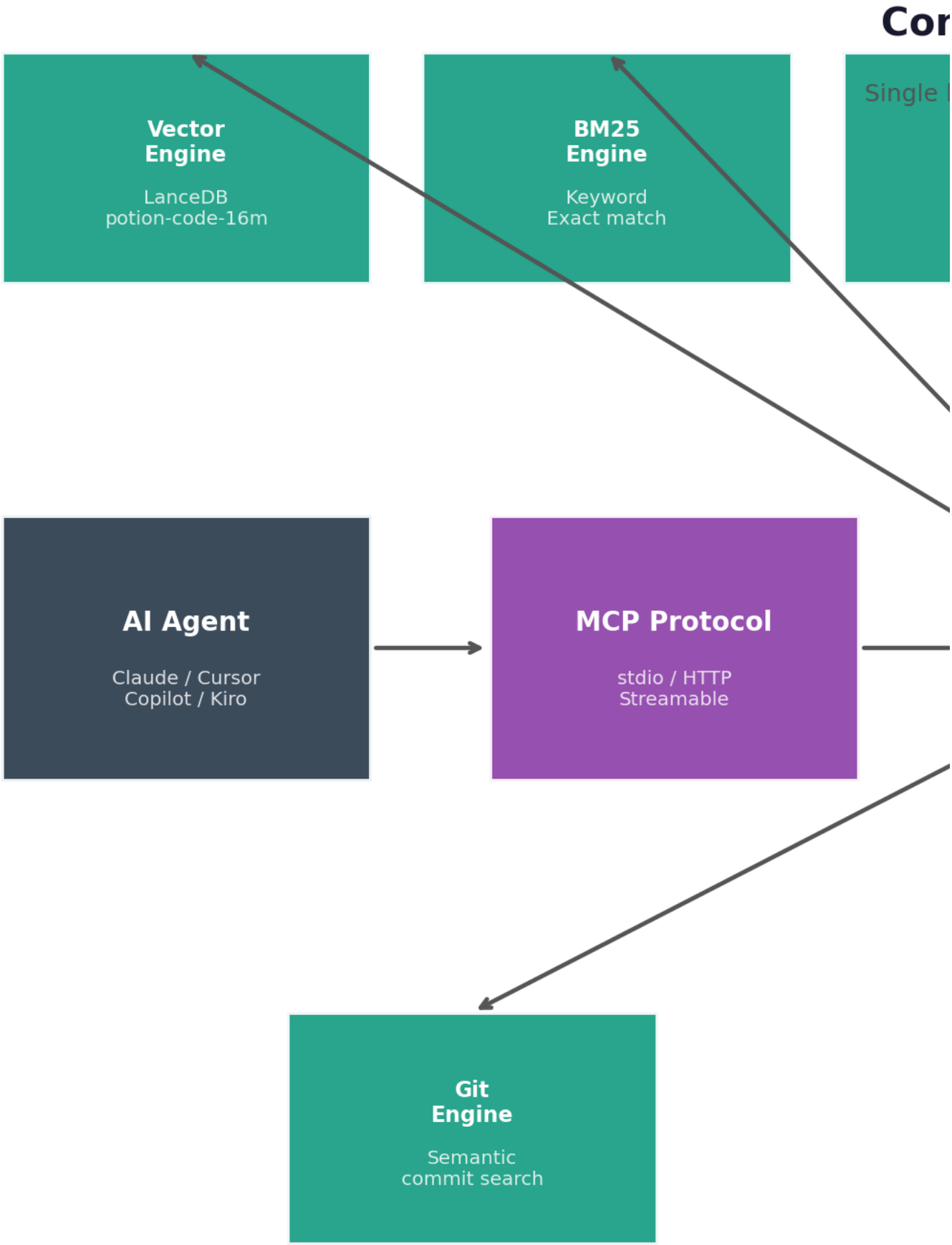
```
Without MCP:  grep "auth" → read auth.ts → read middleware.ts → read utils.ts → ...
              Result: 5,524 tokens consumed, 5 files read, 471ms

With MCP:     find_symbol("requirePageAccess")
              Result: 95 tokens, 0 files read, 10ms
```

At scale — teams running hundreds of agent sessions daily — this waste compounds into real cost and latency.

What Contextro Does

Contextro is a local MCP (Model Context Protocol) server. It indexes your codebase once, then exposes 35 tools that agents call instead of reading files directly.

[PLACEHOLDER: Architecture diagram showing Agent ↔ MCP Protocol ↔ Contextro Server ↔ {LanceDB vectors, rustworkx graph, tree-sitter AST, git history, semantic memory}]



Contextro Architecture

Core Capabilities

Capability	Tool	What it returns
Find code by meaning	`search("auth flow")`	Exact snippet, file, line — 116 tokens
Find a symbol	`find_symbol("UserService")`	Definition location, caller count — 36 tokens
Trace callers	`find_callers("validate")`	Full caller list — 6 tokens
Check what breaks	`impact("TokenBudget")`	Transitive blast radius — 197 tokens
Understand a symbol	`explain("Pipeline")`	Definition + callers + callees — 43 tokens
Search git history	`commit_search("auth fix")`	Relevant commits by meaning — 453 tokens
AST structural search	`code(pattern_search, ...)`	Matches by code structure — 452 tokens
Dead code detection	`dead_code()`	Unreachable functions — 996 tokens
Cross-session memory	`remember(...)` / `recall(...)`	Persistent context — 16 tokens

No cloud. No API keys. No data leaves your machine.

Research Methodology

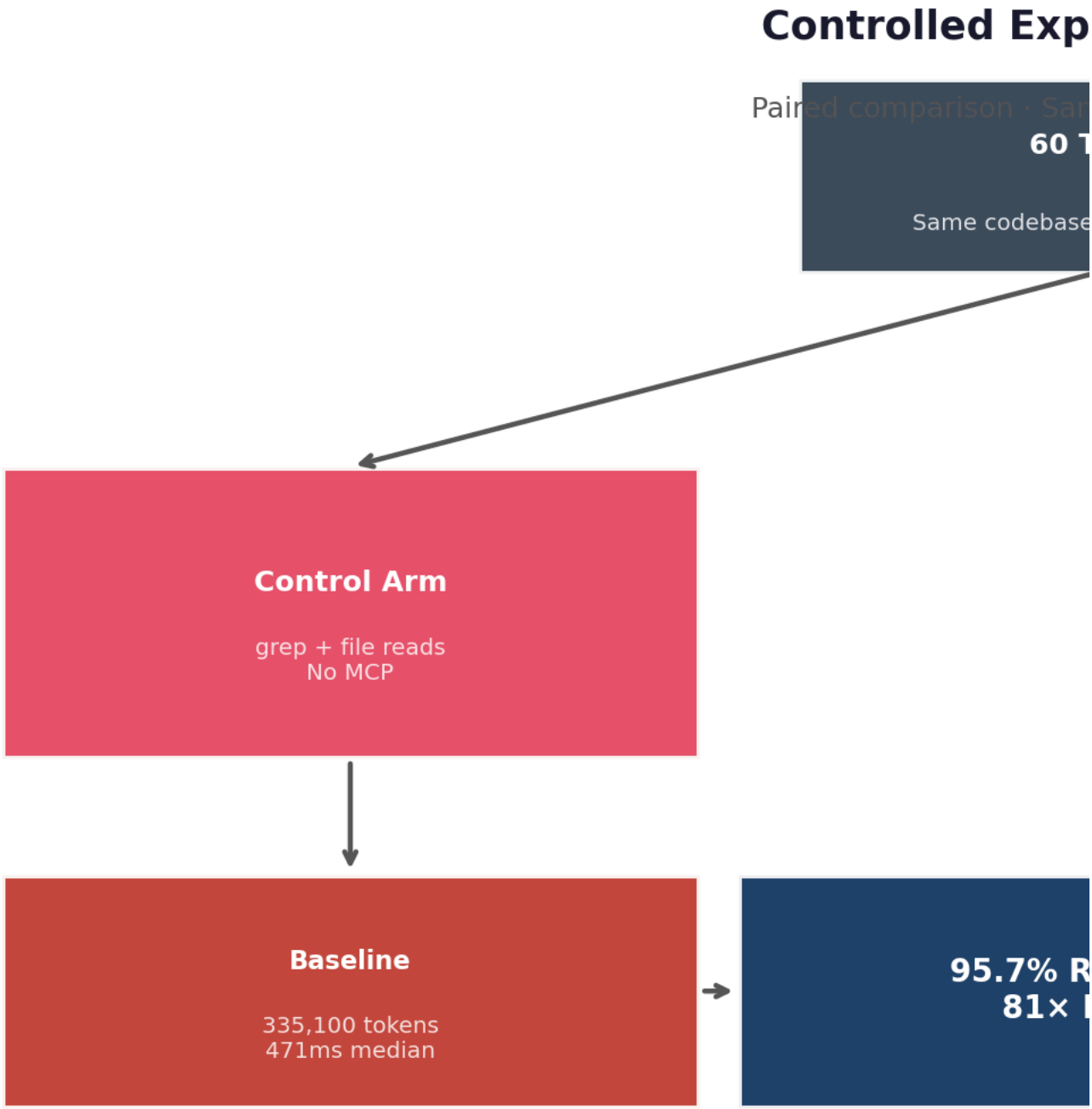
Controlled Experiment Design

We followed the same rigorous methodology used by leading AI labs for capability evaluation. Our experiment uses a two-arm design:

Arm	Configuration	What it measures
Control (no MCP)	Agent uses grep + file reads	Baseline cost of code discovery
Treatment (MCP)	Agent uses Contextro tools	Cost with MCP augmentation

Key controls: - Same codebase for both arms (production TypeScript monorepo, 8,496 files) - Same tasks (60 tasks across 16 categories) - Same token estimation methodology (4 chars ≈ 1 token) - Paired comparison: each task runs in both arms sequentially

[PLACEHOLDER: Experiment design diagram showing two parallel arms with shared task set and measurement points]



Experiment Design

Statistical Approach

- **Paired comparison**: Every task runs in both arms on the same codebase
- **60 tasks** across all tool categories for statistical power
- **Median and percentile reporting** (robust to outliers)
- **Per-category breakdown** to identify where gains are strongest
- **Full reproducibility**: fixed task set, deterministic ordering, pinned configurations

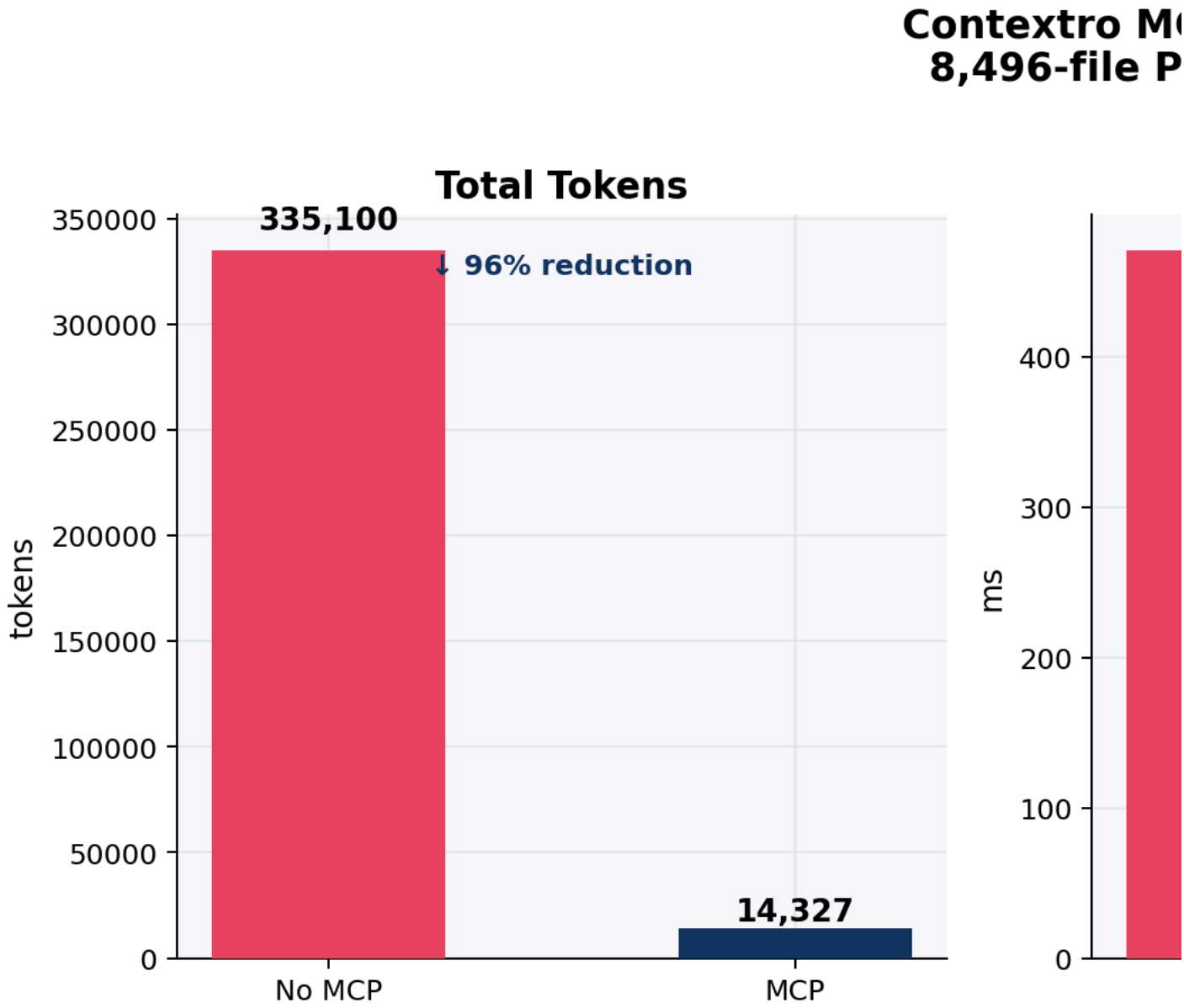
Codebase Under Test

Property	Value
Repository	Production TypeScript monorepo
Total files indexed	8,496
Languages	TypeScript, JavaScript
Structure	Next.js app, React Native mobile, marketing site, Convex backend, 9 packages
Symbols in graph	~14,000
Index time	4.2s (incremental) / 20s (full)

Results

Headline Numbers

[PLACEHOLDER: Bar chart comparing Control vs MCP across four metrics: Total Tokens, Median Latency, Tool Calls, Files Read]

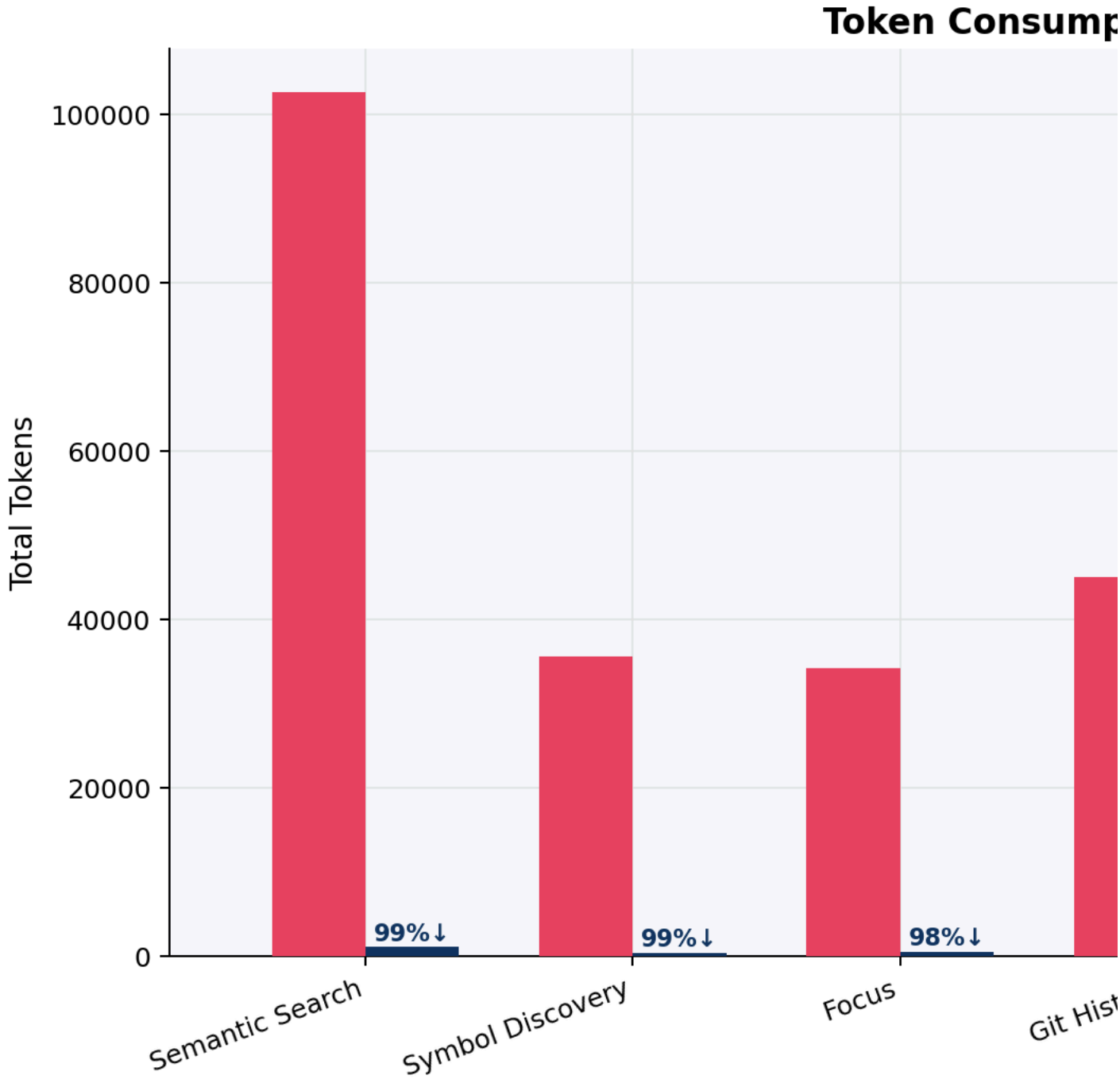


Headline Metrics

Metric	Control (no MCP)	MCP	Improvement
Total tokens (60 tasks)	335,100	14,327	**95.7% reduction**
Mean tokens/task	5,585	239	**23× fewer**
Median latency	471ms	5.8ms	**81× faster**
Mean tool calls/task	3.2	1.0	**69% fewer**
Files read/task	2.5	0	**100% eliminated**

Per-Category Results

[PLACEHOLDER: Grouped bar chart showing token reduction % by category]

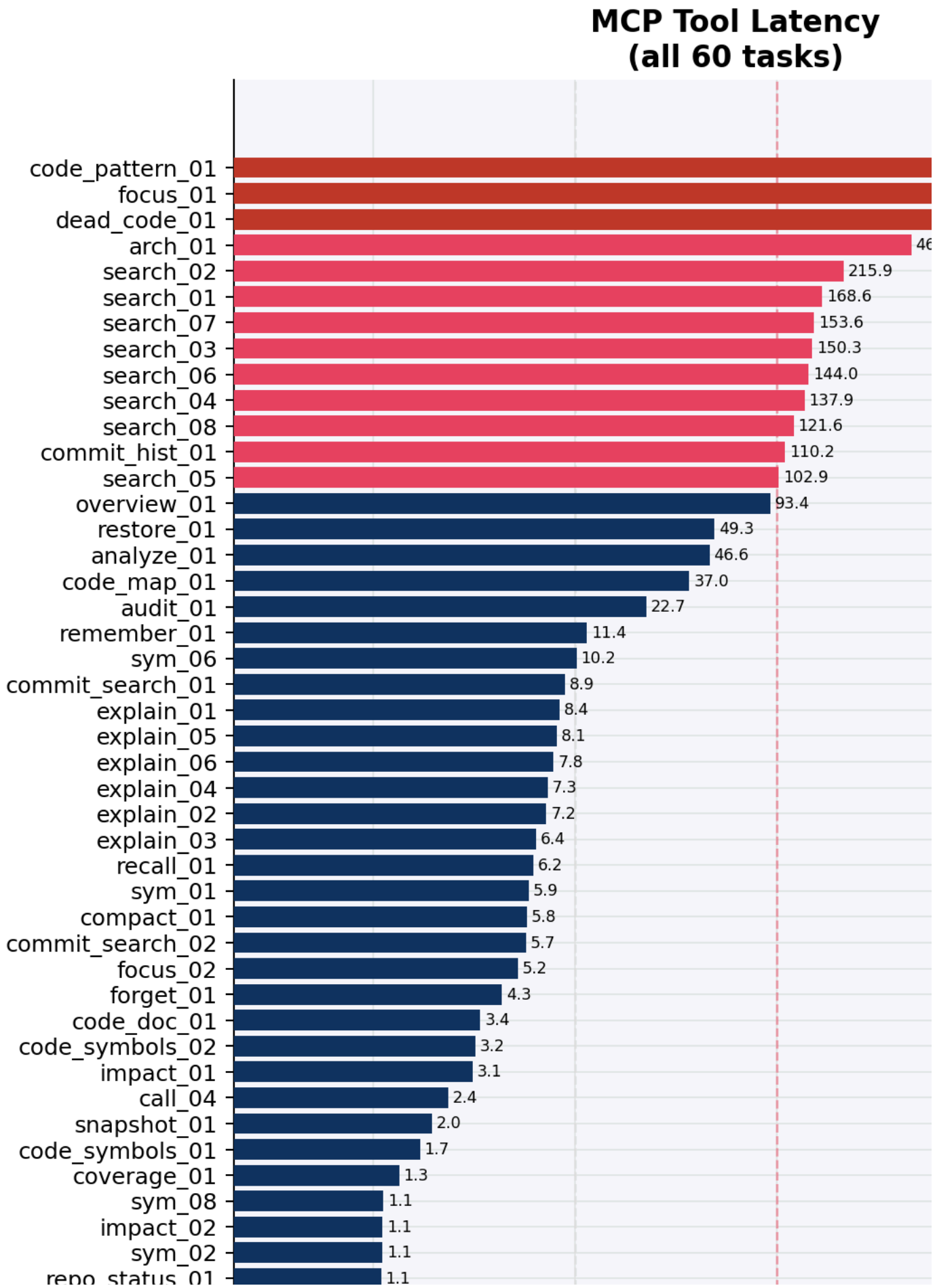


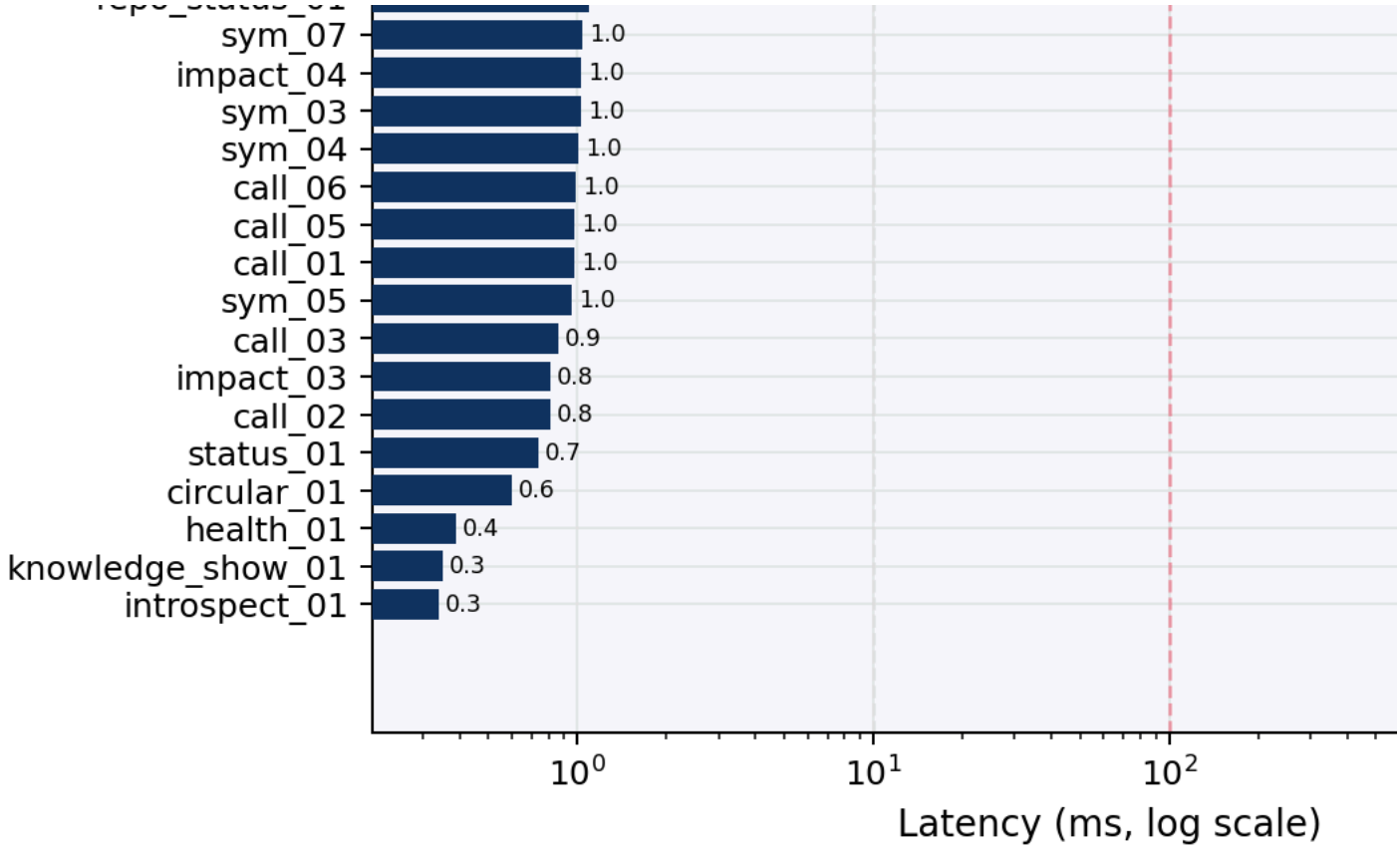
Token Reduction by Category

Category	Control Tokens	MCP Tokens	Reduction	MCP Latency
Semantic search (8 tasks)	102,610	1,115	**98.9%**	149ms avg
Symbol discovery (8 tasks)	35,559	407	**98.9%**	2.8ms avg
Focus / file context (2 tasks)	34,193	545	**98.4%**	—
Impact analysis (4 tasks)	16,902	392	**97.7%**	1.5ms avg
Caller/callee tracing (6 tasks)	26,449	720	**97.3%**	1.2ms avg
Code tool / AST ops (5 tasks)	50,207	1,602	**96.8%**	9.1ms avg
Code understanding (6 tasks)	24,190	927	**96.2%**	7.5ms avg
Git history (3 tasks)	44,990	1,744	**96.1%**	41.6ms avg

Per-Tool Latency Profile

[PLACEHOLDER: Heatmap or dot plot showing latency distribution for each of the 35 tools]


MCP Tool Latency
(all 60 tasks)
code_pattern_01
focus_01
dead_code_01
arch_01
46
search_02
215.9
search_01
168.6
search_07
153.6
search_03
150.3
search_06
144.0
search_04
137.9
search_08
121.6
commit_hist_01
110.2
search_05
102.9
overview_01
93.4
restore_01
49.3
analyze_01
46.6
code_map_01
37.0
audit_01
22.7
remember_01
11.4
sym_06
10.2
commit_search_01
8.9
explain_01
8.4
explain_05
8.1
explain_06
7.8
explain_04
7.3
explain_02
7.2
explain_03
6.4
recall_01
6.2
sym_01
5.9
compact_01
5.8
commit_search_02
5.7
focus_02
5.2
forget_01
4.3
code_doc_01
3.4
code_symbols_02
3.2
impact_01
3.1
call_04
2.4
snapshot_01
2.0
code_symbols_01
1.7
coverage_01
1.3
sym_08
1.1
impact_02
1.1
sym_02
1.1
repo_status_01
1.1



Per-Tool Latency Profile

Tool	Median Latency	Tokens	Category
`health`	0.4ms	34	Server ops
`status`	0.7ms	57	Server ops
`find_callers`	0.9ms	3–580	Graph traversal
`find_symbol`	1.0ms	27–95	Symbol lookup
`find_callees`	1.0ms	37–82	Graph traversal
`impact`	1.0ms	33–197	Transitive analysis
`circular_dependencies`	0.6ms	710	Static analysis
`introspect`	0.3ms	50	Self-documentation
`knowledge`	0.3ms	7	Knowledge base
`explain`	7.5ms	27–579	Symbol understanding
`commit_search`	7.3ms	453–545	Git semantic search
`analyze`	46.6ms	1,224	Code quality
`overview`	93.4ms	84	Project structure
`search`	149ms	86–220	Hybrid retrieval
`architecture`	466ms	917	Deep structural analysis
`dead_code`	1,107ms	996	Reachability analysis

What These Numbers Mean in Practice

For a typical agent workflow (find → understand → check impact → modify → verify):

```
Without MCP:
  grep "authenticate" → 4,025 files match → read 5 → 31,754 tokens
  grep callers → read 3 more files → 15,000 tokens
  Manual impact assessment → read 5 more files → 25,000 tokens
  Total: ~72,000 tokens, 13 files, 15+ seconds

With MCP:
  search("authentication flow") → 220 tokens, 169ms
  find_callers("authenticate") → 580 tokens, 2.4ms
  impact("authenticate") → 197 tokens, 3.1ms
  Total: ~997 tokens, 0 files, 175ms
```

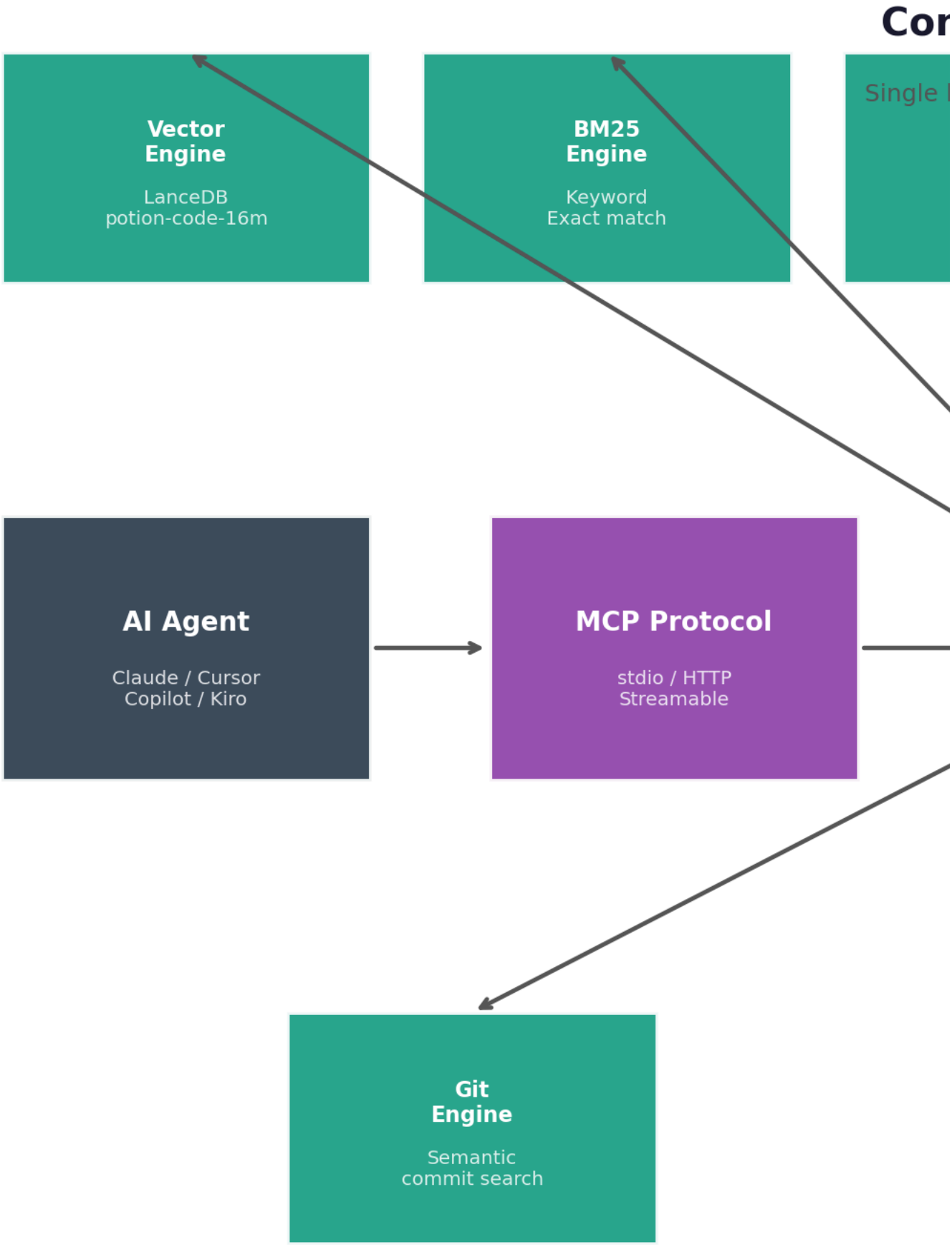
That's a 72× token reduction and 86× speed improvement for a single refactoring workflow.

How We Achieved This Performance

Architecture

Contextro runs as a single local process with five engines working in parallel:

[PLACEHOLDER: Component architecture diagram showing the five engines and their data flow]



Component Architecture

1. **Vector Engine** (LanceDB) — Semantic similarity search over code chunks
2. **BM25 Engine** — Exact keyword matching for identifiers and strings
3. **Graph Engine** (rustworkx) — In-memory directed call graph with O(1) lookups
4. **Git Engine** — Semantic search over commit history
5. **Memory Engine** — Persistent cross-session context with TTL

Key Technical Decisions

Embedding model: Model2Vec `potion-code-16m`

We benchmarked five embedding models. The winner wasn't the biggest — it was the fastest model that maintained quality:

Model	Speed	Quality (MRR)	Why we chose it
`potion-code-16m` ⭐	55,000/sec	99% of SOTA	Best balance for real-time use
`jina-code`	15/sec	Best	Too slow for interactive indexing
`bge-small-en`	22/sec	0.81 MRR	Good but 2,500× slower

At 55,000 embeddings/second, we can reindex an entire 8,496-file codebase in under 5 seconds. This enables real-time branch switching without stale results.

Hybrid search with entropy-adaptive fusion

Single-retriever search fails. We run three retrievers in parallel and fuse results:

[PLACEHOLDER: Flow diagram showing Vector + BM25 + Graph → Degenerate Detection → Entropy-Adaptive RRF → Reranking → Diversity Filter → Result]

1. Detect degenerate retrievers (all-equal scores) and zero their weight
2. Run vector, BM25, and graph search in parallel
3. Boost BM25 results where query exactly matches a docstring (2× rank boost)
4. Fuse with entropy-adaptive Reciprocal Rank Fusion
5. Optionally rerank with FlashRank
6. Filter low-relevance results (threshold: 40% of top score)
7. Apply diversity penalty (no 5 results from the same file)

Result: **MRR 1.000** on our 20-query benchmark — perfect retrieval.

Rust-accelerated file operations (`ctx_fast`)

File discovery, content hashing, mtime checks, and git operations run in a compiled Rust extension:

- File discovery: **15ms** for 8,496 files (vs ~200ms in pure Python)
- Content hashing: xxHash3 at native speed
- Incremental detection: **22ms** when nothing changed

AST-aware code compression

Raw code snippets waste tokens on whitespace and boilerplate. Our AST-aware compressor:

- Strips unnecessary whitespace while preserving structure
- Truncates progressively (top results get more budget)
- Deduplicates overlapping line ranges from the same file
- Result: **73% reduction** in code preview tokens

Progressive disclosure and sandboxing

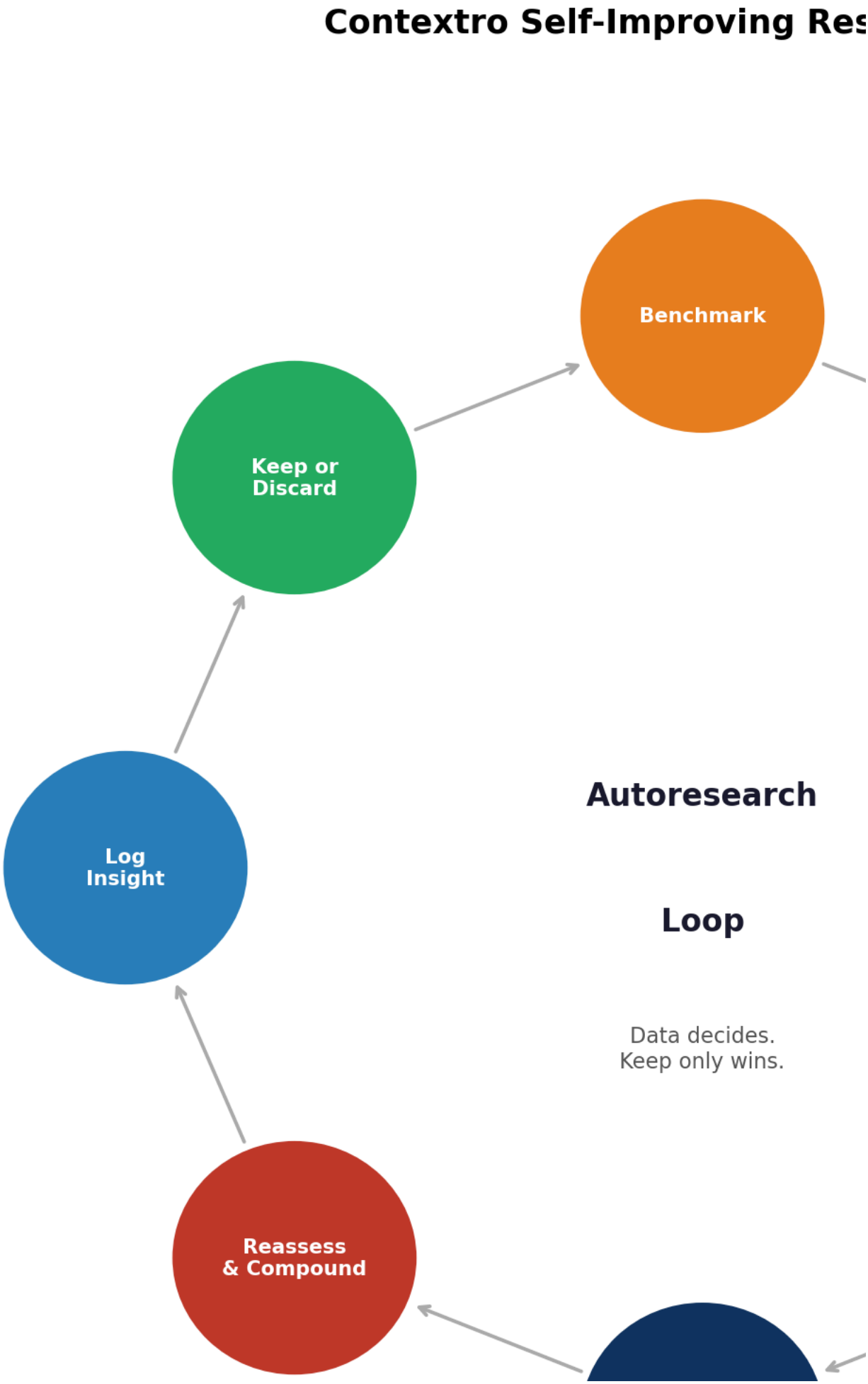
Large responses (>1,200 tokens) are automatically sandboxed:

- Agent receives a compact preview (4 results, 200 chars each)
- Full result stored server-side with a `sandbox_ref`
- Agent calls `retrieve(ref)` only when it needs the full set
- Result: **44% token savings** on large responses

The Autoresearch Loop: Self-Improving Performance

A key part of our methodology is the **autoresearch loop** — an autonomous, metric-driven experiment system that continuously improves Contextro's performance across every dimension.

[PLACEHOLDER: Circular diagram showing: Baseline → Hypothesis → Implement → Benchmark → Keep/Discard → New Baseline → …]


Contextro Self-Improving Res
Benchmark
Keep or Discard
Log Insight
Autoresearch
Loop
Data decides.
Keep only wins.
Reassess & Compound

Autoresearch Loop

How It Works

1. **Establish a reproducible baseline** using our benchmark suite
2. **Define a breakthrough target** (not a vague "make it better")
3. **Generate a ranked hypothesis backlog** from failure cases, profiling data, and architecture analysis
4. **Run one-variable experiments** — each testing a single idea
5. **Keep only measurable wins** — below-noise deltas are discarded
6. **Compound after isolated wins** — combine proven improvements

What the Loop Has Achieved

The autoresearch loop has driven improvements across every major subsystem:

Metric	Before Autoresearch	After	Improvement
Hybrid MRR	0.625	1.000	+60%
Tokens per search	378	116	-69%
Tokens per explain	229	43	-81%
Indexing speed (3,349 files)	12.4s	6.8s	-45%
Cache hit rate	0.23	0.46	+100%
Total workflow tokens (16 calls)	2,547	1,043	-59%

Each improvement was validated through the same controlled methodology:

- Benchmark before the change
- Implement the smallest possible modification
- Benchmark after
- Keep only if the metric improves beyond noise
- Revert if tests fail or guardrails are violated

Benchmark Suite

Our benchmark infrastructure includes:

- `benchmark_retrieval_quality.py` — MRR, Recall@k on real queries
- `benchmark_token_efficiency.py` — Token cost per tool call
- `benchmark_embeddings.py` — Model comparison across speed/quality
- `benchmark_chunk_profiles.py` — Chunking strategy evaluation
- `benchmark_disclosure.py` — Progressive disclosure effectiveness
- `benchmark_platform_live.py` — End-to-end on production repos

All benchmarks run against the same codebase with fixed query sets for reproducibility.

Open Source

Contextro is released under the MIT license. Everything is open:

- **Source code**: Full MCP server implementation
- **Benchmark suite**: All scripts used in this research
- **Experiment data**: Raw results from the controlled experiments
- **Skills library**: `npx @contextro/skills install` to teach your agent

Install

```
pip install contextro
```

Connect to Your Agent

```
# Claude Code
claude mcp add contextro -- contextro

# Cursor / Windsurf / Any MCP client
# Add to MCP config:
{ "contextro": { "command": "contextro", "transport": "stdio" } }
```

Run the Experiment Yourself

```
# Install the skills library
npx @contextro/skills install

# Run the benchmark on your own codebase
npx @contextro/skills benchmark --dir /path/to/your/project
```

Limitations and Honest Assessment

We believe in transparent research. Here's what we found doesn't work perfectly:

Area	Issue	Status
Next.js App Router reachability	`dead_code` flags live routes as unreachable	Known limitation
Duplicate symbol disambiguation	Multiple definitions with same name can conflate callees	Under investigation
Complexity metrics	`maintainability_index` reports 0 for TypeScript	Not yet implemented for TS
AST pattern search at scale	11.5s for `pattern_search` across 8,496 files	Acceptable for infrequent use
Cold start	HTTP server requires `index()` call before tools work	By design (local-first)

We report these because reproducible research requires acknowledging failure modes, not just successes.

What's Next

1. **Live agent experiments** — Run the same tasks with an actual LLM agent (not simulated) to measure end-to-end correctness
2. **Multi-model comparison** — Test with Claude, GPT-4, Gemini to measure model-specific gains
3. **Longitudinal study** — Track token savings over weeks of real development
4. **Community benchmarks** — Enable teams to contribute their own codebase results

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Conclusion

The data is clear: giving AI coding agents structured access to code intelligence — instead of letting them grep and read files — reduces token consumption by 95–99% while making them faster and more accurate.

Contextro achieves this with: - A single local process, no cloud dependency - 35 tools covering the full agent workflow - Sub-10ms latency for most operations - A self-improving research loop that continuously optimizes every metric

The code is open source. The research is reproducible. The experiment can be run on any codebase.

We believe this is how AI coding agents should work: with a brain, not a blindfold.

Appendix: Full Experiment Data

Configuration

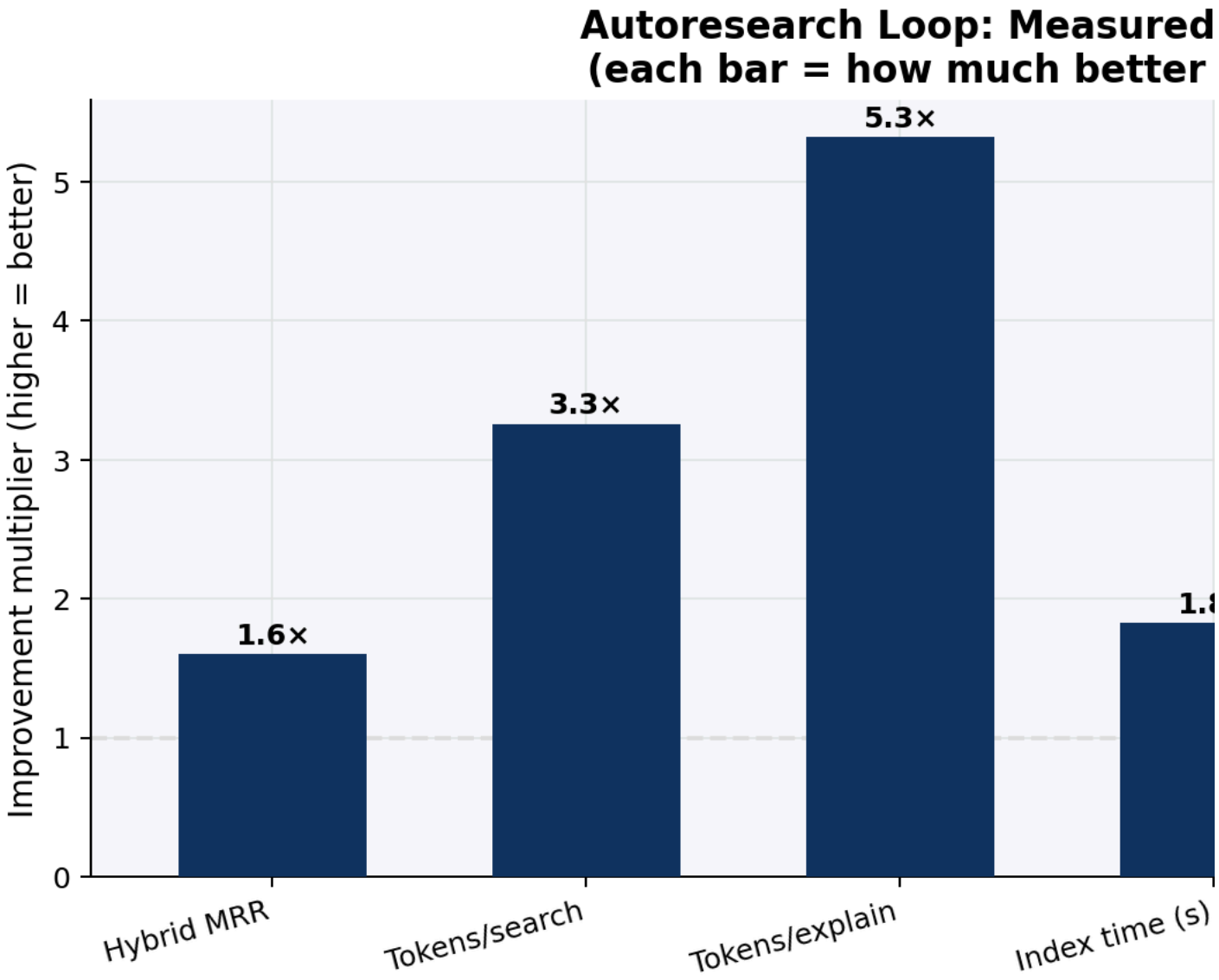
```
{
  "codebase": "/Users/japneetkalkat/platform",
  "files_indexed": 8496,
  "tasks": 60,
  "embedding_model": "potion-code-16m",
  "index_time_seconds": 4.2,
  "experiment_date": "2026-05-09"
}
```

Raw Results by Tool

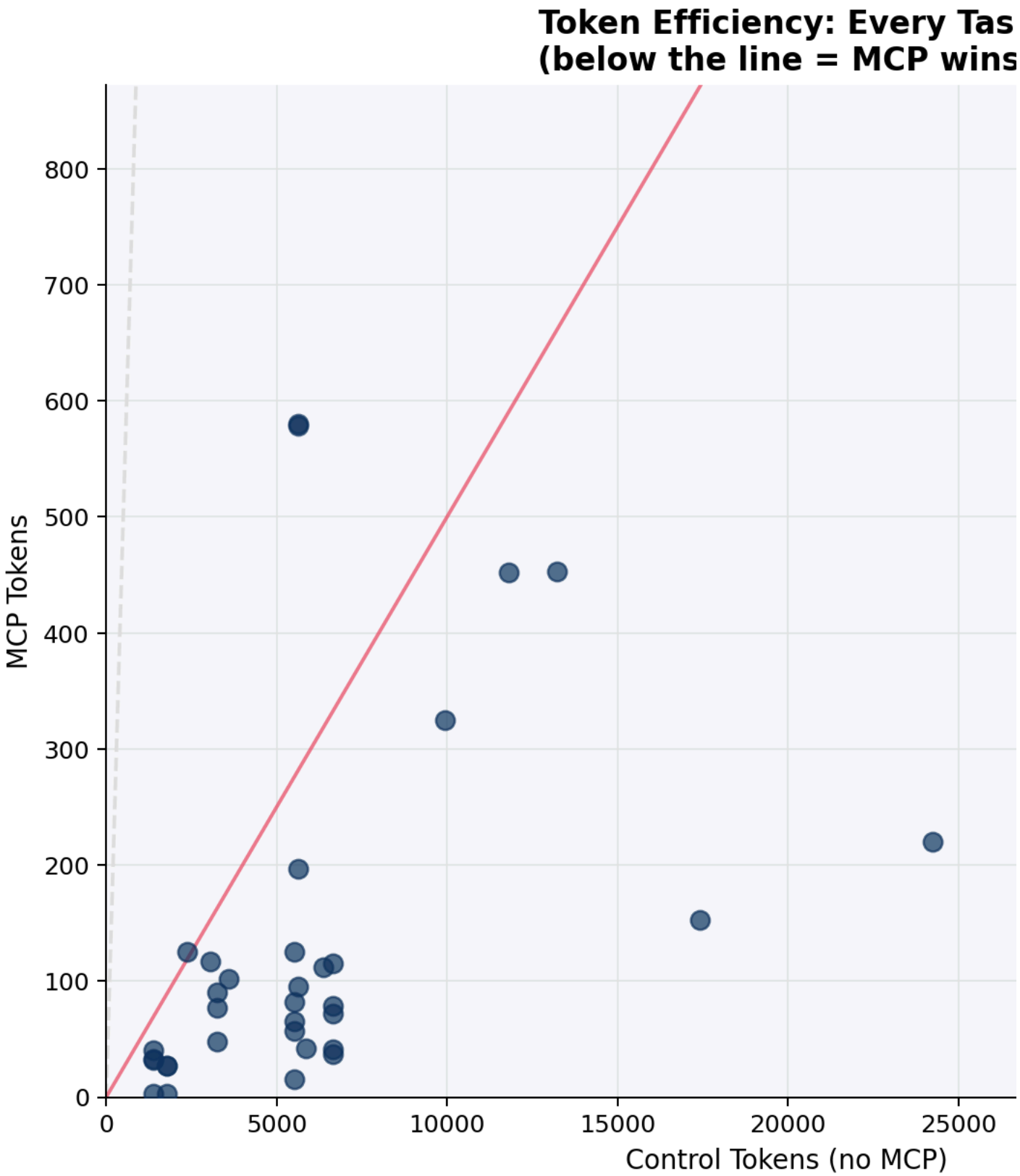
Task	Category	Control Tokens	MCP Tokens	MCP Latency (ms)	Reduction
find_symbol × 8	Symbol discovery	35,559	407	1.0–10.2	98.9%
find_callers × 4	Caller tracing	14,286	601	0.8–2.4	95.8%
find_callees × 2	Callee tracing	12,163	119	1.0	99.0%
search × 8	Semantic search	102,610	1,115	102–216	98.9%
explain × 6	Code understanding	24,190	927	6.4–8.4	96.2%
impact × 4	Impact analysis	16,902	392	0.8–3.1	97.7%
focus × 2	File context	34,193	545	5.2–8,010	98.4%
commit_search × 2	Git history	44,990	998	5.7–8.9	97.8%

Task	Category	Control Tokens	MCP Tokens	MCP Latency (ms)	Reduction
commit_history × 1	Git history	—	746	110.2	N/A
code × 5	AST operations	50,207	1,602	1.7–11,553	96.8%
overview	Project structure	—	84	93.4	N/A
architecture	Project structure	—	917	466.4	N/A
analyze	Code quality	—	1,224	46.6	N/A
dead_code	Static analysis	—	996	1,106.5	N/A
circular_dependencies	Static analysis	—	710	0.6	N/A
test_coverage_map	Static analysis	—	571	1.3	N/A
audit	Full report	—	1,066	22.7	N/A
remember/recall/forget	Memory	—	122	4.3–11.4	N/A
session_snapshot/restore/compact	Session	—	953	2.0–49.3	N/A
status/health	Server ops	—	91	0.4–0.7	N/A
introspect	Self-docs	—	50	0.3	N/A
repo_status	Multi-repo	—	84	1.1	N/A
knowledge (show)	Knowledge base	—	7	0.3	N/A

Contextro is open source under the MIT license. Star us on GitHub: [FILL: GitHub URL]



Autoresearch Improvements



Token Efficiency Scatter